



Earn₹s

Coding & Web Development (Lecture 02)

How internet works?

Summary Notes

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Sim Wifi
↓
Internet
Tio, Airtel

Telecome company / ISP (Internet service provider)

SIM card - a tiny chip inside your phone that identifies you to a telecom company, like an ID card. It's how Jio/Airtel/Vi know "this phone belongs to this customer."

SIM Aadhar card , address , photo



9876543210

Telecom company / ISP (Internet Service Provider) - a company that owns physical infrastructure (towers, cables, equipment) and sells you access to the internet. Jio, Airtel, Vi, BSNL, ACT Fibernet - these are all ISPs. This is the company you're actually paying when you "recharge."

Telecome Company / ISP (Internet Services Provider)

Jio, Airtel, VI , BSNL

Infrastructure (Mobile towers , cables)

Mobile tower - those tall metal/antenna structures you see around your city. Your phone talks to the nearest tower wirelessly using radio waves (the same basic idea as a walkie-talkie, just far more advanced).

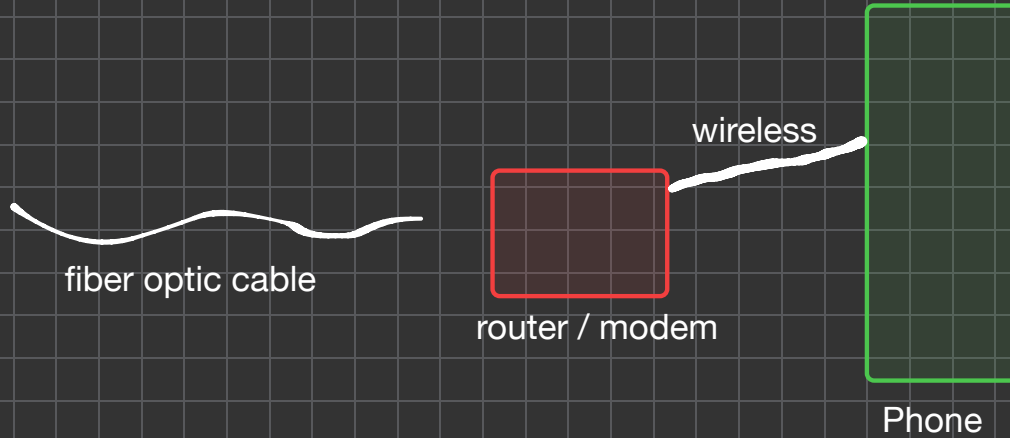
Data - the actual information traveling back and forth: a webpage, a video frame, a WhatsApp message. Measured in MB (megabytes) and GB (gigabytes). When your recharge plan says "2GB/day," that's the amount of data you're allowed to send/receive that day before the company slows you down or charges extra.

Recharge - literally a payment to your ISP for a fixed amount of data + calling minutes, valid for a fixed number of days. You are not "buying internet" out of thin air - you're paying a real company that owns real physical equipment to let your phone use their network.

Broadband - a wired (cable/fiber) internet connection straight into your house, instead of going through a mobile tower. Usually used for home WiFi.

Router / Modem - the box at home (often with blinking lights) that takes the broadband connection coming into your house and converts it into a WiFi signal your phone/laptop can connect to.

Broadband wired

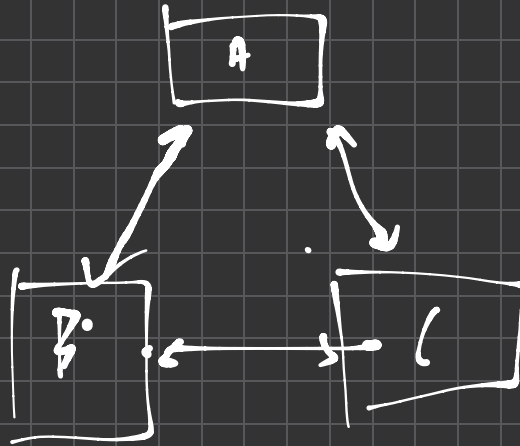


Wifi - > internet

Connected , No Internet

Internet

Network ???



Network - two or more computers connected so they can talk to each other. Your home WiFi is a small network. Your college's WiFi is a bigger network.

4 types of network

1. PAN (Personal area network)
2. LAN (Local area network)
3. MAN (Metropolitan area network)
4. WAN (Wide area network)

MAN (Metropolitan Area Network) — multiple LANs conn

1. PAN

Mobile <----> smart watch

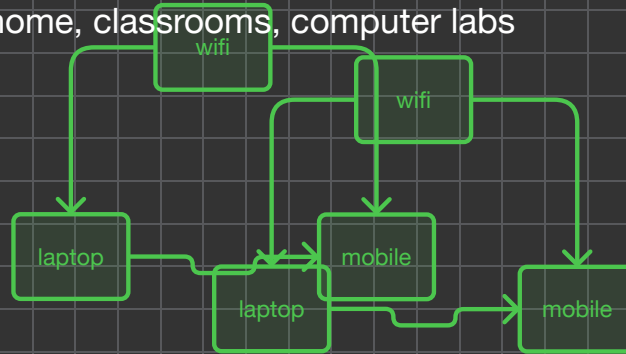
Mobile <-----> earbuds

PAN (Personal Area Network) - the smallest possible network, covering just a few meters around one person. Example: your phone connected to Bluetooth earphones, or a smartwatch synced to your phone.

LAN -> local area network

home, classrooms, computer labs

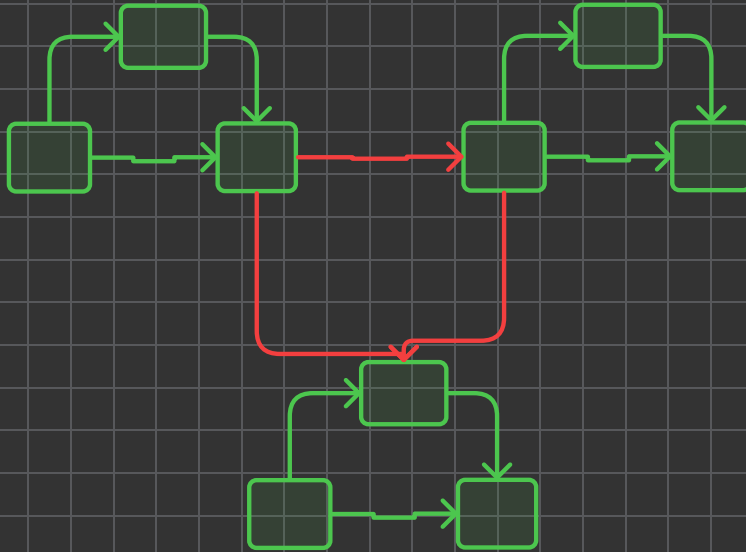
home, classrooms, computer labs



LAN (Local Area Network) - devices connected within one limited physical place: one house, one classroom, one office floor, one school building. Example: every phone/laptop connected to your home WiFi router is one LAN. Your school's computer lab, all wired together, is also a LAN. Range: roughly one building, often under 1 km.

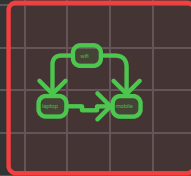
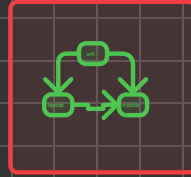
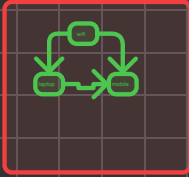
MAN -> Metropolitan area network

Big university, cities



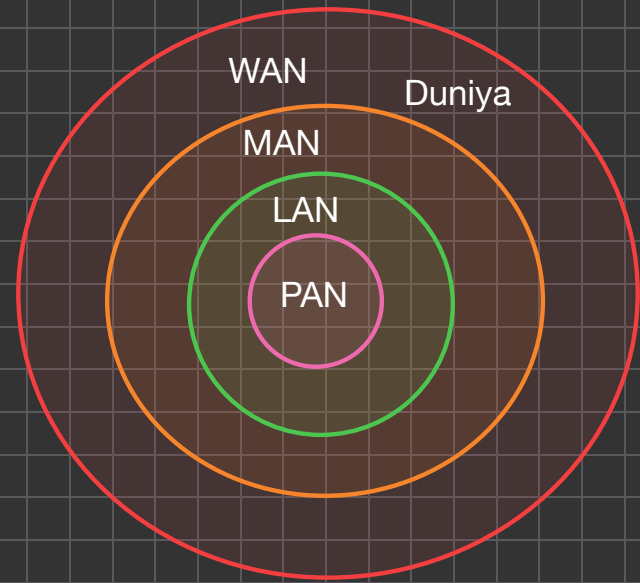
MAN (Metropolitan Area Network) - multiple LANs connected together across a whole city or a large campus with many buildings. Example: a cable TV company's network covering an entire city; a university connecting its 10 different department buildings together; a city-wide public WiFi project.

Multiple departments



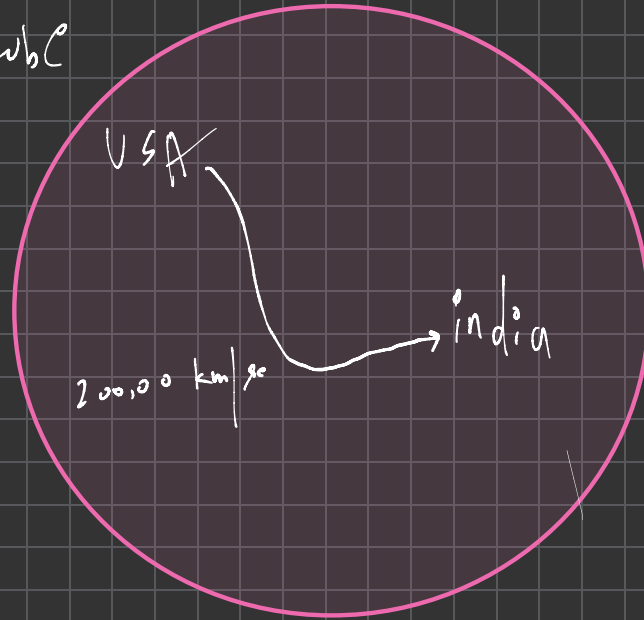
WAN - > Wide area network

states, countries, continents

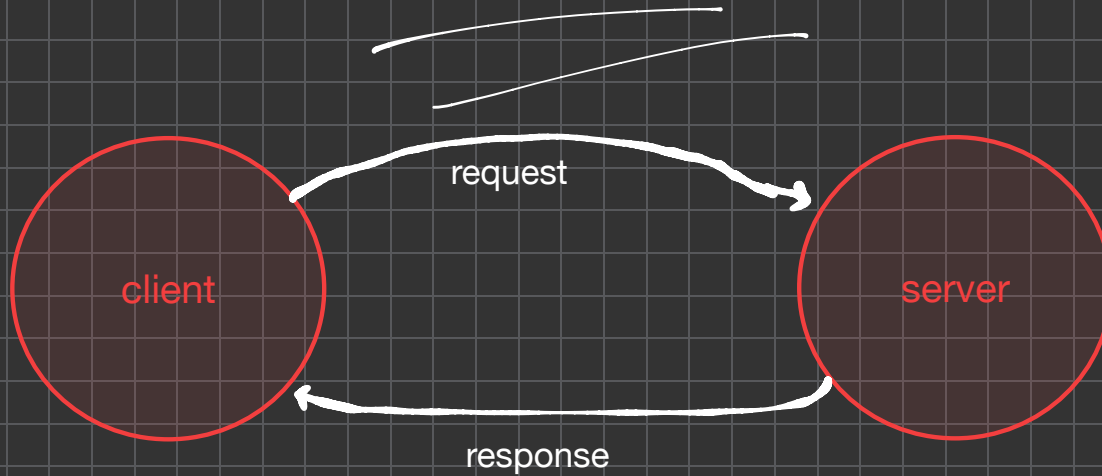


WAN (Wide Area Network) - a network spanning a huge area: multiple cities, states, countries, or even the entire planet. Example: your ISP's nationwide backbone connecting every city it serves is a WAN.

youtube



client server model



Client - the one who asks for something. Your browser (Chrome, Safari, etc.) is a client. Your phone's Instagram app is a client.

Server - the one who has the thing and sends it back. A server is just a powerful computer that sits "always on," waiting for requests, somewhere in a data center.

Request - the message the client sends ("hey, give me the YouTube homepage").

Response - the message the server sends back (the actual HTML/video/data).

IP Address

internet protocol

machine / device address ----> ip

google.com ?
domain

Varsha -> 975927345
Subh -> 8137861356
Ankit -> 762387635
Basudev -> 972349807
Unnati -> 87134712530

IP Address (Internet Protocol Address) - a unique number assigned to every device connected to the internet, so other devices know where to send data. Looks like: **142.250.183.78**

Windows: open Command Prompt, type ipconfig
Mac: open Terminal, type ifconfig or ipconfig
getifaddr en0

DNS

Domain name system

```
youtube.com ---> 196.128.1.3  
google.com ---> 196.128.1.8  
instagram.com ---> 196.128.1.5  
x.com ---> 196.128.1.24  
xyz.com ---> 196.128.1.23  
pw.com ---> 196.128.1.4
```

DNS (Domain Name System) - a giant, distributed phonebook that converts a domain name (like youtube.com) into the actual IP address of the server that owns it.

Domain name - the human-readable website name, like amazon.com.

Domain Name Server - a special server whose entire job is to answer the question "what IP address does this domain point to?"

```
nishantsaini@Nishants-MacBook-Pro ~ % nslookup google.com
Server:      2401:4900:50:9::7dd
Address:     2401:4900:50:9::7dd#53

Non-authoritative answer:
Name:   google.com
Address: 192.178.193.139
Name:   google.com
Address: 192.178.193.102
Name:   google.com
Address: 192.178.193.138
Name:   google.com
Address: 192.178.193.100
Name:   google.com
Address: 192.178.193.113
Name:   google.com
Address: 192.178.193.101

nishantsaini@Nishants-MacBook-Pro ~ % nslookup youtube.com
Server:      2401:4900:50:9::7dd
Address:     2401:4900:50:9::7dd#53

Non-authoritative answer:
Name:   youtube.com
Address: 142.251.220.78
```

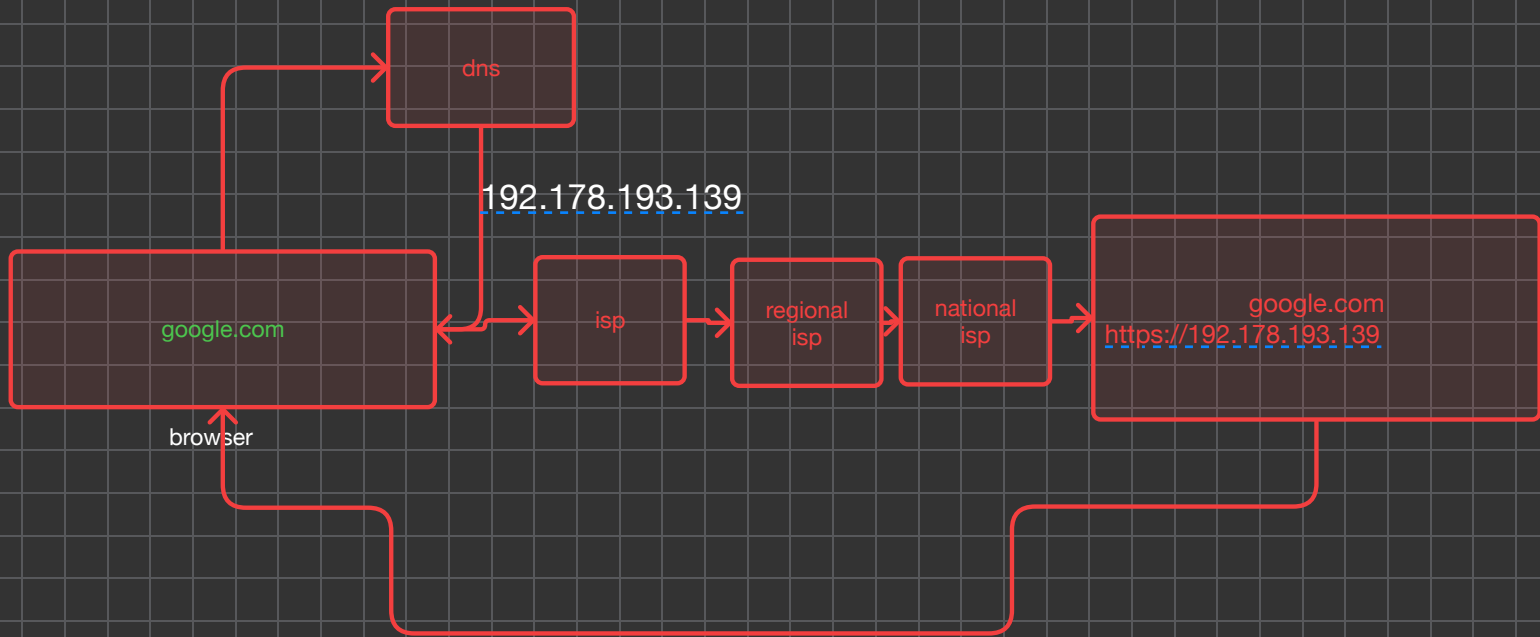
Walk through the actual sequence on the board, step by step, like a story:

You type youtube.com and hit Enter.

Your browser asks a DNS server: "What's the IP address for youtube.com?"

DNS replies: "It's 142.250.183.78."

Now your browser knows exactly which computer on Earth to talk to.



nslookup

window -> ipconfig

mac- > ifconfig

~~IPV4~~ ☒ ~~IPV6~~

code

HTML

IPv4

IPv4 (Internet Protocol version 4) - the original, older addressing format. Made of 4 numbers (each 0-255) separated by dots, like 142.250.183.78. Because each number is built from 32 binary bits total, IPv4 can only create about 4.3 billion unique addresses in total.

Internet Protocol Version 4

32 bits

X.X.X.X

X = 0-255

eg 3.4.5.7

ipconfig

☒ cmd
ipconfig

4.3 billions

8 billions

IPv6

IPv6

IPv6 (Internet Protocol version 6) - the newer format built specifically to solve that shortage. It looks completely different - longer, and uses letters and colons instead of just dots, like:
2401:4900:1c20:abcd:0000:0000:0000:1a2b

128 bit Hexadecimal

ab12 : de98 : — : — : — : — : — : —

2401:4900:1c20:abcd:0000:0000:0000:1a2b

0-9 , a-f

340 undecillion

IPv6